

Light-triggered, Ultrafast Gate Driver Empowered by GaN Buffer for High Power SiC Module

Annoy Kumar Das*, Shahabuddin Khan[†], Zheyu Zhang[‡], Fred Wang[§], and Hua Bai[¶]

^{*,†,‡}Department of Electrical, Computer, and Systems Engineering, Rensselaer Polytechnic Institute, Troy, NY, USA

[§]CURRENT Engineering Research Center, University of Tennessee, Oak Ridge National Laboratory, Knoxville, TN, USA

[¶]CURRENT Engineering Research Center, University of Tennessee, Knoxville, TN, USA

Email: dasa7@rpi.edu*, khans10@rpi.edu[†], zhangz49@rpi.edu[‡], fred.wang@utk.edu[§], hbai2@utk.edu[¶]

Abstract—Owing to their inherent isolation and high noise-immunity, optically controlled gate drivers are desirable for deriving disruptive technologies for high-voltage, high-frequency, and high-power applications. However, in light of the emerging wide bandgap (WBG) semiconductor devices, improvement of the FoMs (figure of merits) of optically controlled gate drivers, such as power capacity and switching time vs. optical input power, is desirable. This paper proposes a light-triggered gate driver (LTGD) consisting of a single low-power InGaAs photodiode (PD) and low-voltage GaN buffer transistors to achieve ultrafast switching as well as capable driving current for high power SiC module. First, concept of the proposed LTGD and its circular economy compatible design is presented. Next, modeling and experimental characterization of the key components, including PD and GaN buffer transistors, are presented. Thereafter, simulation in LTspice considering detailed components' models is established and modes of operations are discussed. Finally, the proposed light-triggered gate driver is verified experimentally. The results demonstrate 57/38 ns gate voltage rise/fall time with > 50 A sinking/sourcing current for 3.3 kV, 624 A SiC power module with over 52 nF of equivalent input capacitance.

Index Terms—GaN buffer, LTspice, optically controlled gate driver, photodiode, SiC module.

I. INTRODUCTION

High-voltage (HV), high-power semiconductor devices are usually controlled by an auxiliary power electronic (PE) circuit, i.e., gate driver, to switch between high blocking voltage and high conduction current. In recent times, optically driven power devices and/or gate drive arrangements have gained traction, owing to their promise to enable electromagnetic interference (EMI) immunity, an ideal isolation between the power and control stages, a reduced triggering delay, and a simplified control for device stacking [1]. Conventional optically controlled power devices, such as photoconductive semiconductor switch (PCSS) and light-triggered thyristor (LTT), lead to trade-off between the optical power and switching speed. This is overcome by optically activated gate driver, which uses optical devices only in the low-power driver and enables energy-efficient optical control of unipolar devices [2], [3]. However, as illustrated in Figs. 1(a)–(b), the results reported in literature encompass at best a blocking voltage up to 1.5 kV and a continuous drain current of 3 A [4], only, which is practically not adequate to efficiently drive high power SiC module. Therefore, with the mW optical power in-

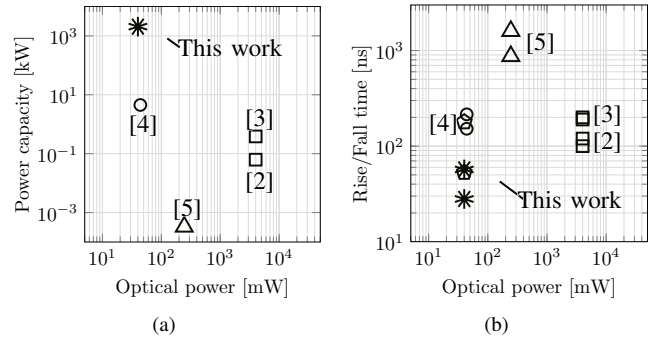


Fig. 1. Comparison of prior arts and this work based on (a) power capacity vs. optical power, and (b) rise/fall times vs. optical power, respectively; FoMs related to this work are shown using * and $\circ$ markers, which are derived from LTspice simulation and experimental results, respectively.

put, further research in the quest for improving the gate driver output driving capacity and then reducing the switching time is desirable. Such solutions are particularly attractive for deriving disruptive technologies for grid transmission and renewable energy applications, such as high power module with multiple high voltage power semiconductor chips connected in parallel.

In this work, we propose a light-triggered, ultrafast and noise-immune gate driver (LTGD), which employs only a single PD triggered by optical signal (from controller), and utilizes a “Two-stage” circuit in conjunction with GaN buffer transistor to amplify its gate driving current capability. The proposed LTGD shown in Fig. 2 is inherently isolated, yields minimized traces/wires thereby reducing electromagnetic interference (EMI) and expediting SiC triggering, and hence suitable for driving high voltage and/or high power SiC modules, and achieving voltage scalability of the main power device. Therefore, the proposed light-triggered gate driver has inherent potential to support the the voltage scalability of the main power device, which is critical in deriving disruptive technologies for grid resilient infrastructures.

The remaining contents of this paper are organized as follows. In Section II, the working principle and salient features of the proposed light-trigger gate driver are briefly discussed. In Section III, detailed-component level modeling of InGaAs PD in LTspice and pulsed based characterization of GaN buffer transistors across outdoor temperature ranging from -50°C

to 125 °C are provided. In Section IV, we demonstrate the proposed LTGD successfully driving > 50 A to achieve fast switching of 3.3 kV, 624 A SiC module having an aggregate input capacitance of 52.3 nF, and report its gate-source voltage rise and fall time limited to 57/38 ns, measured respectively from the rising edge of the turn ON and turn OFF light pulses. These results of the proposed LTGD are among the best performances of optically controlled gate drive arrangements reported in literature for WBG power devices.

II. OVERVIEW OF THE PROPOSED LIGHT-TRIGGERED ULTRAFAST GATE DRIVER EMPOWERED BY GAN BUFFER

Fig. 2 shows the functional block diagram of the proposed LTGD; its salient features are detailed below.

1) *Optical Isolation*: It is fully isolated through light with no need for external switching mode power supply, enabling a significant EMI reduction, and eliminating its influence on the control signals. For example, power over fiber (PoF) approach is one potent solution by utilizing readily available ‘off-the-shelf’ optical power converter (OPC).

2) *Noise-Immunity*: It is highly noise-immune due to its compactness and utilizing negative driving voltages during turn-off operation of SiC module.

3) *Minimum Number of Photodiode Used*: The proposed design utilizes only a single PD to achieve the turn ON and turn OFF operations of 3.3 kV 624 A SiC module, totaling an input capacitance of $C_{tot} = 52,560$ pF. This is a significant improvement in comparison to the results previously reported in literature [4], which utilizes two PDs triggered by complementary optical signals for switching a single power transistor of 1.5 kV blocking voltage and having $C_{iss} = 610$ pF.

4) *“Two-Stage” Architecture*: The proposed LTGD achieves fast actuation and high gate current driving capability by adopting to an intermediate “Two-stage” circuit:

a) *Stage-I Circuit*: This interfaces the output of low optical input power PD using an amplifier thus providing easy access to typical usage with transistor-transistor logic (TTL) logic. Using a zener diode based clamping circuit, the unipolar (+5 V/0 V) pulsed output of PD is then converted into an equivalent bipolar pulse width modulated (PWM) signal (± 5 V), which is used as the designated input signal for the next stage. Note that the proposed LTGD utilizes isolated power rails: VCC1/GND1 for the fiber optic (FO) receiver module, and VCC2/GND2 for the subsequent gate drive circuit, respectively; These isolated power rails are derived in the Stage-I circuit using combinations of linear regulator, isolated dc-dc converter having very low isolation capacitance (> 3 pF), and high gain boost converter in SON package, thus achieving compact design, low parasitics effects and fast transient response. Individual components are carefully sized to improve the overall efficiency of power conversion.

b) *Stage-II Circuit*: This comprises a phase-leg of GaN and pMOS transistors, biasing, clamping and control circuits enabling turn ON and turn OFF operations of the SiC module using a single PD. The purpose of biasing circuits is to derive the negative driving voltage during turn OFF, thus facilitating

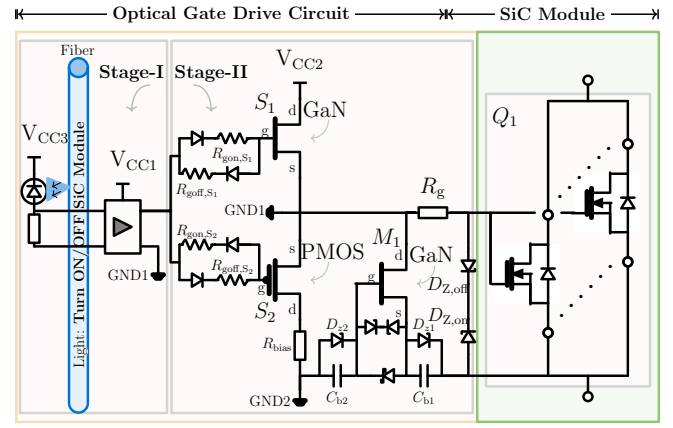


Fig. 2. A schematic representation of the proposed light-triggered gate driver (LTGD) comprising an intermediate “Two-stage” circuit derived using a single low-power photodiode (PD), biasing and control circuits, and low voltage GaN buffer transistors to boost the gate current driving capability and achieve fast response, and capable of driving high power 3.3 kV 624 A SiC module.

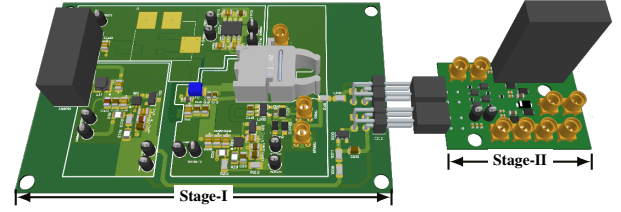


Fig. 3. Proposed LTGD modeled in Altium Designer shows the functional integration of individual PCBs for stage-I and stage-II circuits, thereby illustrating the circular economy compatible design adopted in this work.

fast gate discharge operation. Zener diode based protection circuit is incorporated to clamp the gate voltage of the main SiC module to +20 V/−4 V during turn ON and turn OFF operations. The individual circuit components are rated such that only one dc power rail of VCC2 is adequate to supply the different functional stages in Stage-II circuit during turn ON and OFF operations, thus simplifying the power supply design incorporated in Stage-I circuit.

5) *In-built Shoot-Through Protection*: It is desirable to avoid any sustained short-circuit in the phase-leg, comprising high-side and low-side transistors (GaN) and S_2 (p-channel MOSFET (pMOS)), respectively. In conventional scheme, such as [4], the requisite deadband can be realized via the light pulses for ON and OFF operations derived from the controller. In comparison, the proposed LTGD utilizes a single PD and single optical input from the controller. Hence, the desirable deadband is implemented in an indirect manner by selecting different switching speeds for the complementary transistors S_1 and S_2 by appropriately sizing their turn-on and turn-off gate resistances respectively. This allows to fully utilize the pulsed current capability of GaN buffer transistor without undue thermal stress and switching loss, and also support peak gate driving current capability for the SiC module.

6) *Towards Circular Economy Compatibility*: proposed LTGD has potentials of being an integral part of disruptive

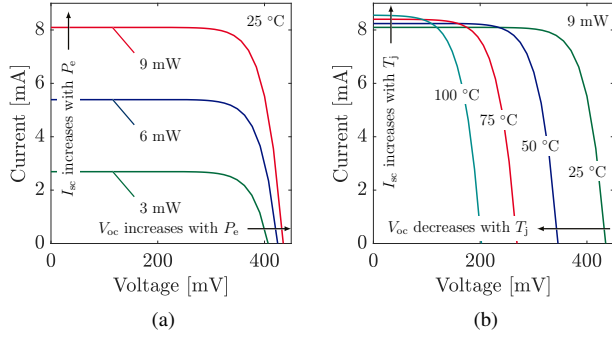


Fig. 4. I-V characteristics of PD modeled in LTspice under different operating conditions (a) optical input power P_e is varied in steps at fixed temperature $T_j = 25^\circ\text{C}$, and (b) operating temperature is changed in steps at constant irradiance level $P_e = 9\text{ mW}$.

HVDC/HVDC technologies for grid resilient infrastructures, such as reliable oil-immersed ultra-compact modular valve with optical control (ROUTER) design proposed in [6]. Therefore, the circularity and lifetime assessment of the proposed LTGD design should be analyzed and evaluated based on a database of ecodesign criteria, such as the ecodesign for sustainable products regulation (ESPR) as envisaged by the EU Ecodesign Directive [7]. To achieve progress toward a circular economy in power electronics, discarded products, modules, or components must be recovered, which mandates the implementation of end-of-life (EoL) strategies. In order to maximize the high-value circularity efficiency of the light-triggered gate driver, EoL choices and constraints are taken into account early on in the design phase itself. Since disassembly is a known bottleneck in successfully implementing EoL practices, ‘design to enable disassembly’ feature is carefully integrated in the design strategy of the proposed LTGD. Fig. 3 illustrates that its modular and scalable architecture enhances the gate-driver’s reconfigurability and disassemblability, its reuse in other product families, as well as the product’s maintenance, and serviceability. Furthermore, modularity inherent to the LTGD encompasses both functional and structural approaches; in addition, due consideration is given to achieve circularity by grouping components based on their similar reliability levels, material values, failure rates, and lifetime value, thus enhancing their circularity value. For example, the “off-the-shelves” available optical components, such as OPC, FO receiver module, etc., usually support an operating temperature range of -40°C to $+85^\circ\text{C}$; however, the local ambient temperature in the vicinity of SiC modules used in high power and high voltage valve designs can easily exceed the upper temperature threshold for optical components, at which their performance is generally guaranteed by the manufacturer. In the proposed LTGD design, the optical components are therefore integrated in Stage-I circuit, which being physically away from the hot-spots, ensures their reliable operation. On the other hand, Stage-II circuit comprises the power electronic components, such as GaN transistors, zener and schottky diodes, etc., which can support much higher threshold for the operating/ambient temperature, and therefore be placed in

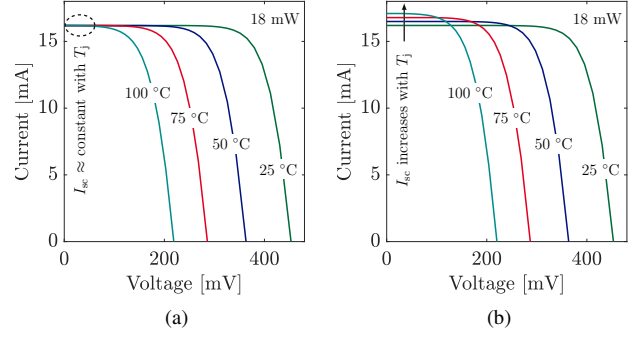


Fig. 5. Effect of responsivity temperature coefficient of PD modeled in LTspice (a) $\alpha = 0$, and (b) $\alpha = 0.075\%$, respectively. This indicates that the short-circuit current i_{sc} is more sensitive to the change in operating temperature T_j , which is aptly modeled in this study.

close proximity of the SiC module, ensuring reduced stray parasitics and reliable electrical performance. Therefore, attention is paid on the design and life cycle management (LCM) of high-failure rate components of the proposed LTGD to further mitigate potential negative effects on the environment. In addition, this will also help reach sustainable development goal, responsible consumption and production, as well as achieve circular economy compatibility [8]–[10].

III. MODELING AND EXPERIMENTAL CHARACTERIZATION OF KEY COMPONENTS OF THE PROPOSED LTGD

A. Temperature-Dependent Modeling of PD in LTspice

Photodiodes (PDs) have been widely used for optical sensing and communication while relatively less attention has been paid in case of the power application. Therefore, few datasets and models are available which a designer can utilize readily. To overcome this limitation, a detailed component-level model adopting fast-response InGaAs photodiode is built in LTspice incorporating its optical, electrical as well as thermal interactions and inter-dependencies, details of which are given in Appendix A. The equivalent circuit of a PD shown in Fig. 13 is modeled in LTspice based on the system parameters of a fiber coupled high-speed InGaAs p-i-n photodiode [11]. Accordingly, the I-V characteristics are derived with respect to different operating conditions, as shown below.

1) *Input Optical Power P_e* : In this simulation, P_e is varied in steps while the temperature is kept constant ($T_j = 25^\circ\text{C}$). Accordingly, the I-V characteristics shown in Fig. 4(a) are obtained. This indicates that the open-circuit voltage V_{oc} is proportional to the log of irradiance levels P_e while the short circuit current I_{sc} increases linearly with P_e .

2) *Operating Temperature T_j* : In this simulation, the optical input power is held constant at $P_e = 9\text{ mW}$, while T_j is varied in steps. The resultant I-V characteristics shown in Fig. 4(b) indicates that V_{oc} decreases more dominantly whereas I_{sc} increases slightly with increasing operating temperature.

3) *Responsivity Temperature Coefficient α* : In this simulation, the effect of α on the I-V characteristics of a PD is studied. Results for $\alpha = 0$ are shown in Fig. 5(a), implying that for a given illumination level, I_{sc} is very nearly insensitive to

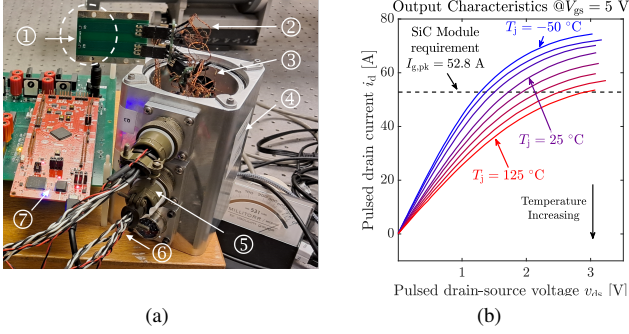


Fig. 6. (a) Experimental platform for characterizing GaN buffer transistor EPC2014 over wide temperature range. Experimental results obtained at different junction temperatures (b) output characteristics @ $V_{gs} = 5\text{ V}$. Measured I-V characteristic clearly indicates that the selected GaN buffer transistor EPC2014 can support the required driving current $> 50\text{ A}$ over wide operating temperature ranging from -50°C to 125°C , while the on-state resistance $R_{ds(on)}$ increases from 24.41 mΩ to 57.35 mΩ.

any change in temperature. However, this is rectified while the effect of α is incorporated in the PD model built in LTspice. Results shown in Fig. 5(b) depicts that at an optical input power level of 9 mW, I_{sc} increases from 8.10 mA at $T_j = 25^\circ\text{C}$ to 8.56 mA at $T_j = 100^\circ\text{C}$, indicating a relative increase of 5.68% in I_{sc} . Nature of these derived I-V characteristics agree well with those provided in the datasheet of photodiodes, and validates the modeling accuracy of PD model built in LTspice.

B. Experimental Characterization of GaN Buffer Transistor

The GaN buffer transistor is an integral part of the proposed LTGD, serving as the buffer transistor, and plays a crucial role in achieving fast turn ON and turn OFF operations of high power SiC module. To ensure feasibility of the selected EPC2014 GaN, its pulsed characterization is performed over wide temperature ranging from -50°C to 125°C in steps of 25°C . Fig. 6(a) shows the experimental testbench indicating its various components, such as ① the device under test (DUT), ② power cable (copper), ③ signal cable (phosphor bronze), ④ vacuum enclosure, ⑤ sealed connector, ⑥ I/O signals connected to the curve tracer, and ⑦ auto selection matrix board; relevant details about the temperature-dependent characterization platform, e.g. in-built functionalities and automation, co-thermal optimization methodology for lead design at cryogenic temperatures, etc. can be found in [12]–[14].

Key findings of the experimental characterization are as follows. The selected 40 V/10 A (60 A_{peak}) GaN buffer transistor (EPC2014) offers chip-scale footprint (around $1.7\text{ mm} \times 1.1\text{ mm}$) and small input capacitance ($C_{iss} = 220\text{ pF}$) thus helping achieve fast and strong driving capabilities with minimized parasitics. The experimentally measured output characteristics of EPC2014 are depicted in Fig. 6(b), which stipulate that at the rated gate-source voltage $v_{gs} = 5\text{ V}$ (adopted for the design of phase-leg in Stage-II circuit of LTGD), the chosen GaN buffer transistor can support the required gate driving current of more than 48 A over wide temperature ranging from -50°C to 125°C , thus validating the selection of EPC2014 GaN for the LTGD design.

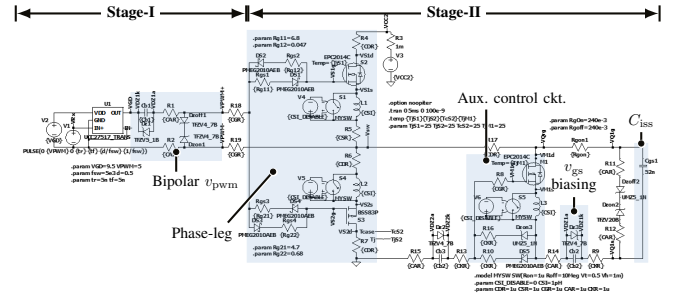


Fig. 7. The light-triggered gate driver modeled in LTspice with different functional stages highlighted, such as bipolar gate pulse generation, phase-leg comprising high-side GaN and low-side pMOS transistors, auxiliary control circuit consisting of GaN transistor M_1 , biasing circuits, and main SiC module Q_1 , emulated by an input capacitance of $C_{iss} = 52\text{ nF}$.

IV. MODEL VERIFICATION OF THE PROPOSED LTGD

A. Simulation Results in LTspice

Fig. 7 shows the proposed LTGD modeled in LTspice using detailed component-level models of GaN, pMOS, Schottky and zener diodes, etc. The various functional stages of the LTGD are discussed as follows. Stage-I circuit comprises a zener diode and bias capacitor-based circuit, which converts the unipolar output v_{Rx} of FO receiver module into an equivalent bipolar pulse waveform v_{pwm} . This serves the following purpose: v_{Rx} being TTL-compatible, cannot drive the complementary pMOS transistor S_2 ; therefore, v_{pwm} of bipolar nature ($\pm 4.7\text{ V}$) is derived using bias capacitor and back-to-back zener diode-based circuit for controlling the phase-leg of Stage-II circuit. The positive and negative voltage levels of v_{pwm} are adjusted depending on the gate characteristics of selected GaN and pMOS transistors. Both turn on and turn off gate resistances are used in the phase-leg to achieve optimal switching speeds of the complementary transistors S_1 and S_2 , thus avoiding any potential shoot-through in the phase-leg. Stage-II circuit also comprises an auxiliary control circuit using GaN transistor M_1 , which facilitates fast turn OFF operation of the SiC module. The 3.3 kV, 624 A SiC module is emulated by an equivalent input capacitance of $C_{iss} = 52\text{ nF}$. Operation of the proposed LTGD is discussed next.

1) *Turn ON Operation:* The relevant waveforms during the turn ON operation are shown in Fig. 8. When light pulse is received, output of FO receiver module turns on the high-side GaN transistor S_1 via action of the bipolar pulse generation circuit; on the other hand, v_{pwm} being positive, turns off the low-side pMOS transistor S_2 . This effectively extends the isolated power supply $VCC2$ to the gate terminals of SiC module, and v_{gs} begins to build up. The biasing capacitors C_{b1} , C_{b2} , and the zener diodes across them, D_{z1} , D_{z2} , respectively, are sized such that for $VCC2 = 32\text{ V}$, at steady-state, a gate voltage of $+20\text{ V}$ is achieved across the main SiC module. During this period, the auxiliary GaN transistor M_1 is turned off on account of the forward voltage drop across Schottky diode D_{sbd} . As indicated in Fig. 8(c), the rise time of v_{gs} is estimated to be 39.6 ns; from the rising edge of turn ON light pulse, the total time to fully turn ON the SiC

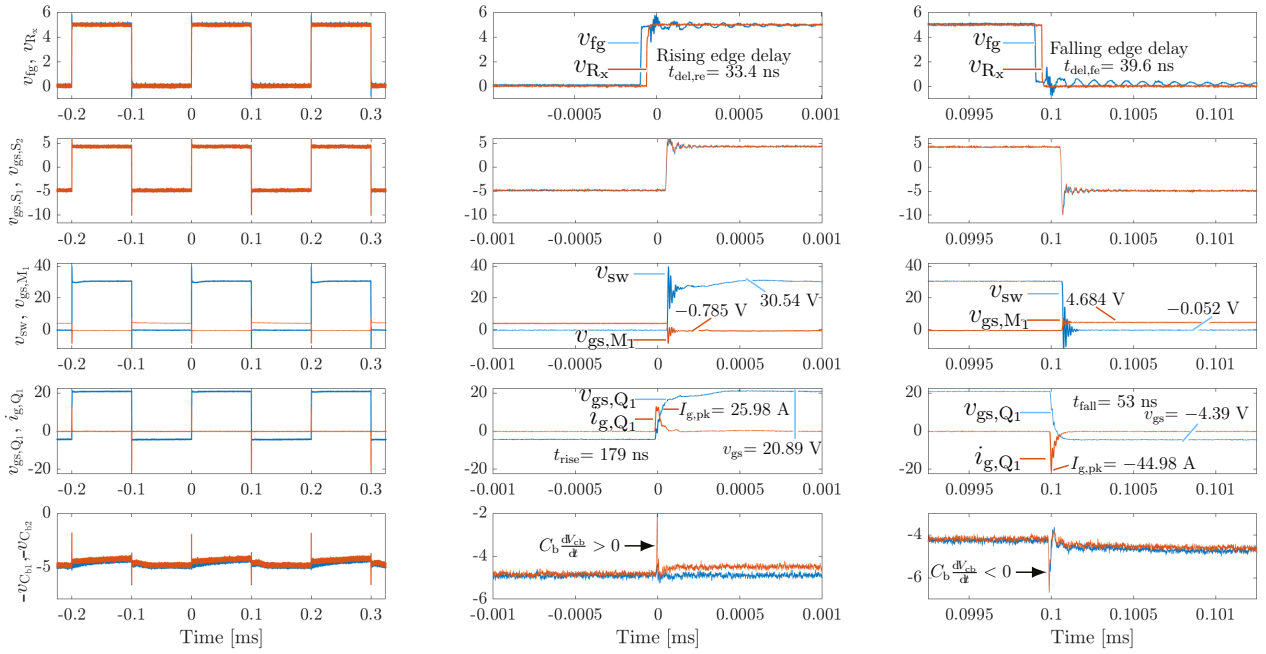


Fig. 10. Experimental results are shown indicating key waveforms during steady-state as well as during turn ON and turn OFF operations. The measured results demonstrates a 45 A_{peak} current sinking capability of the designed light-triggered gate driver.

B. PCB Schematic and Layout Design of “Two-Stage” LTGD

Based on the simulation model built in LTspice discussed above, the schematic and layout design of the proposed light-triggered gate driver is carried out. Fig. 11 illustrates the PCB layout, indicating the various functional stages of the light-triggered gate driver. To simplify the testing and debugging process, the transmitter section is also populated in the same PCB as that of the LTGD; in practice, the transmitter sections will be designed separately and located physically away from the gate driver section; the control/actuating signals for turn ON and turn OFF operations will be communicated from the transmitter to the receiver units through fiber optic (FO) communication using multimode 62.5/125 fiber patch cables, the respective ports for which are indicated in Fig. 11. Careful design practices are followed during PCB layout design.

a) For example, an optimal power loop design strategy [15] is adopted by placing the decoupling capacitors in close proximity to the high-side GaN device, with the positive input voltage terminals located next to its drain connections. This improved layout technique reduces the loop size, has magnetic field self-cancellation, has inductance that is independent of board thickness, is a single-sided component PCB design, and yields high efficiency for a multilayer PCB.

b) The line grid array (LGA) package-based EPC2014 GaN buffer transistor has its drain and source terminals extending on both sides of the device to minimize drain and source inductance. These traces then are connected immediately down to all the subsequent PCB layers through a number of parallel and interleaved vias to reduce the loop inductance as much as possible. Interleaving of the vias with current flowing in an opposing direction reduces magnetic energy storage and

helps generate magnetic field cancellation, resulting in reduced eddy and proximity effects, reduced ac conduction losses. The interleaved arrangement of the vias for source and drain terminals of EPC2014C GaN transistors are shown in Fig. 11.

c) Stage-II circuit is designed using surface mounted device (SMD) components with 0402 package footprint; this compact design keeps the stray parasitics' effects to minimum.

d) For measurement purposes, MMCX connectors are used to reduce the impact of stray parameters during sensing. Kelvin connection scheme is adopted while populating traces for the measurement of sensitive signals (e.g., gate-source voltage, output of the fiber optic T_x/R_x modules, etc.) to improve the measurement accuracy and avoiding any undesirable influence by other circuit components.

C. Experimental Results

Working principle of the proposed LTGD is also verified with the help of experimental results shown in Fig. 10. The designed PCBs for Stage-I and Stage-II circuits are shown in Fig. 11. The power quality of the isolated power supply rails derived in the Stage-I circuit, i.e., VCC1 = 9.5 V and VCC2 = 32 V, are found to be satisfactory. To support PWM operation by the gate driver, the circuit components of LTGD (and PCB traces) carrying the peak gate charging current (> 50 A) are adequately sized to avoid thermal stress under continuous operation. Key waveforms, such as a) input and output signals v_{fg} and v_{R_x} of the FO transmitter and receiver modules, respectively, b) gate-source voltages of the complementary transistors S_1 and S_2 , and M_1 , v_{gs,S_1} , v_{gs,S_2} , and v_{gs,M_1} , respectively, c) switching node voltage v_{sw} , and d) the gate-source voltage v_{gs,Q_1} and current i_{g,Q_1} of Q_1 , representing the

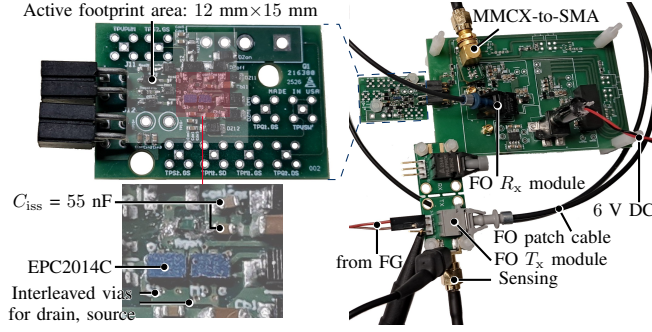


Fig. 11. Assembled “Two-stage” circuit of the proposed LTGD is shown indicating the Stage-I circuit comprising the FO receiver and power supply stages, and Stage-II circuit comprising the phase-leg, GaN transistors, control and biasing circuits. Input to the FO transmitter module is derived from a function generator (FG), not shown here. Zoomed view of Stage-II circuit shows the additional capacitances added to achieve gate $C_{iss} = 52$ nF.

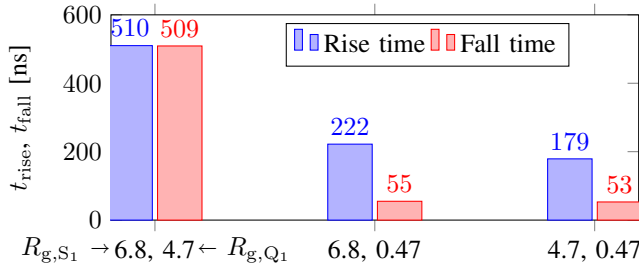


Fig. 12. Experimental results showing the effect of gate turn on resistances (shown along x-axis in Ω) on the rise and fall times of v_{gs} of SiC module.

SiC module, are recorded and shown in Fig. 10 under steady-state conditions, as well as during turn ON and turn OFF conditions. In this setup, gate resistance $R_g = 0.47 \Omega$ is used for the main SiC module having an input capacitance of $C_{iss} = 55$ nF (this is emulated by populating additional capacitors $C_{gs,e1}$ and $C_{gs,e2}$, as illustrated in Fig. 11). The measured results demonstrate that a) 26 A_{peak} current sourcing is noted during turn ON operation, whereas b) 45 A_{peak} current sinking is observed during turn OFF operation, thus validating the high current sourcing/sinking capability of the proposed light-triggered gate driver. Fig. 12 depicts that the turn off speed of $v_{gs,Q1}$ is most sensitive to the gate resistance $R_{g,Q1}$, whereas its turn on speed is dependent on the gate turn on resistance $R_{g,S1}$ of also. For $R_{g,S1} = 4.7 \Omega$, and $R_{g,Q1} = 0.47 \Omega$, rise and fall times of $v_{gs,Q1}$ of SiC module are found to be 179 ns and 53 ns, respectively. Therefore, by tuning these gate turn on and turn off resistances, LTGD performance can be further improved in terms of shorter rise and fall times, and faster switching speed of SiC module.

V. CONCLUSIONS

This work proposes a LTGD consisting of low-power InGaAs PDs and low-voltage GaN buffer transistors to achieve ultrafast switching with capable driving current for high power SiC module. Simulation results with detailed component modeling in LTspice corroborates that the proposed technique of employing fast-speed InGaAs PDs, fast GaN with high pulsed

current capability, and an intermediate “Two-stage” biasing and control circuit, helps augmenting the power capacity versus optical input power substantially from 0.102 kW/mW (best performance reported in literature) to 2.578 kW/mW (this work). This enables to achieve ultralow rise/fall time of the gate-source voltage of SiC module limited to 57/38 ns, measured respectively from the rising edge of the turn ON and turn OFF light pulses. Efficacy of the proposed LTGD is demonstrated while switching a 3.3 kV 624 A SiC power module with a total input capacitance of $C_{iss} = 52,560$ pF, 86 $\times$ higher than previously reported results (a single power transistor of 1.5 kV blocking voltage and $C_{iss} = 610$ pF). Experimental results validate that the selected GaN buffer transistor can support pulsed driving current of > 50 A over wide temperature ranging from -50°C to 125°C .

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APPENDIX A

MODELING OF PHOTODIODE IN LTSPICE

Photodiodes (PDs) are semiconductor devices responsive to high energy particles and photons. By absorption of photons or charged particles, PDs generate a flow of current in an external circuit in proportional to the incident light. An electrical equivalent circuit of PD comprises a current source I_{ph} in parallel with an ideal diode V_D , junction capacitance C_j , shunt resistance R_{sh} , and followed by a series resistance R_s , as shown in Fig. 13. Therefore, the output current I_o of a PD connected to an output load of R_L is given by

$$I_o = I_{ph} - I_D - I_{sh} \quad (\text{A.1a})$$

where, I_{ph} is the photocurrent generated by the incident light of optical input power of P_e at a given wavelength λ . Considering the variations on the characteristics of responsivity on account of change in temperature, I_{ph} is calculated as

$$I_{ph} = R_{\lambda,1} (1 + \alpha (T_2 - T_1)) \cdot P_e \quad (\text{A.1b})$$

where, $R_{\lambda,1}$ is the optical responsivity of PD at temperature T_1 ; α is the temperature coefficient of responsivity; T_2 is elevated temperature. The diode current I_D is calculated as

$$I_D = I_{sat} \left(e^{\frac{qV_A}{k_B T_j}} - 1 \right) \quad (\text{A.1c})$$

where, I_{sat} is the reverse saturation current of PD representing thermal-generated free carriers which flow through the junction (dark current); V_A is the applied bias voltage; T_j is the junction temperature in Kelvin; $q = 1.602 \times 10^{-19}$ C is the electron charge; and $k_B = 1.38 \times 10^{-23}$ J/K is the Boltzmann constant. I_{sh} is current through the shunt resistance R_{sh} , which is the slope of I-V curve of PD at the origin. Utilizing these

analytical expressions in (A.1), the open circuit voltage V_{oc} and short-circuit current I_{sc} of a PD at a given level of illumination are derived as follows.

1) *Open-Circuit Voltage V_{oc}* : If a PD is kept open then an open-circuit voltage V_{oc} is generated at its terminals with positive polarity at the anode [cf. Fig. 13(a)]. V_{oc} is derived by substituting the boundary condition $I_o = 0$ in (A.1a); in addition, note that voltage across the diode is the open-circuit voltage i.e., $V_D = V_{oc}$. Further simplification yields

$$V_{oc} = \frac{k_B T}{q} \ln \left(\frac{I_{ph} - \frac{V_{oc}}{R_{sh}}}{I_{sat}} + 1 \right) \quad (A.2a)$$

Assuming I_{sh} is negligible due to large value of R_{sh} , and since I_{sat} increases exponentially with respect to temperature T , the open-circuit voltage V_{oc} is a) inversely proportional to temperature T , and b) proportional to the natural log of I_{ph} .

2) *Short-Circuit Current I_{sc}* : If a PD is shorted at its terminal i.e., $R_L = 0$, this gives rise to a short-circuit current I_{sc} , emanating from the anode. This is calculated by substituting the boundary condition $I_o = I_{sc}$; in addition, note that the voltage across the diode is given by $V_D = I_{sh} R_{sh} = I_{sc} R_s$. Further simplification results in the short-circuit current as

$$I_{sc} = \frac{1}{1 + \frac{R_s}{R_{sh}}} \left(I_{ph} - I_{sat} \left(e^{\frac{q R_s I_{sc}}{k_B T}} - 1 \right) \right) \quad (A.2b)$$

Assuming $R_{sh} \gg R_s$, this indicates that a) I_{sc} increases linearly with the light intensity. b) In addition, I_{sc} is dependent on the operating temperature through the resistivity temperature coefficient α of a PD [cf. (A.1b)]; for instance, if α is positive at a given wavelength, then I_{sc} increases with the operating temperature, which is captured in Fig. 5 in Section III-A3.

3) *LTspice Model*: An equivalent model of a PD shown in Fig. 15 is built in LTspice. Using this, the I-V characteristics of PD are derived, which are discussed in Section III-A.

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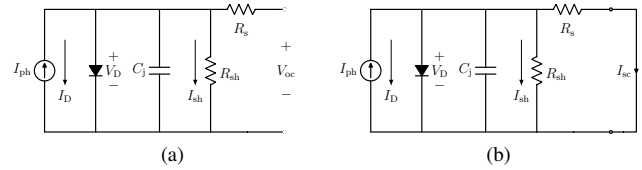


Fig. 13. Deriving (a) open-circuit voltage V_{oc} , and (b) short-circuit current I_{sc} in terms of the circuit parameters of a PD for constructing I-V characteristics.

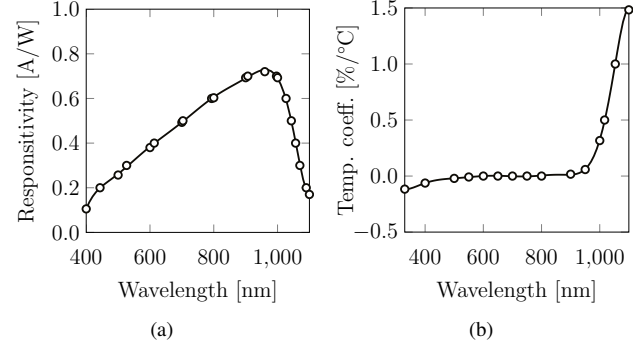


Fig. 14. (a) Typical responsivity vs. wavelength, and (b) responsivity temperature coefficient vs. wavelength characteristics of a PD [16]. Curve-fit in MATLAB is done based on discrete data obtained from datasheet. This imbues the sensitivity temperature characteristics of a PD in its LTspice model.

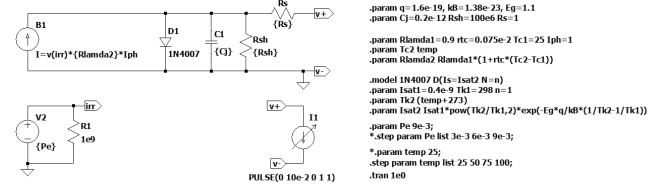


Fig. 15. photodiode model built in LTspice; temperature dependency is modeled through responsivity R_λ , and saturation current I_{sat} of PD.

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